



Do metacommunities vary through time? Intermittent rivers as model systems

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Abstract

Aim: Metacommunity assembly mechanisms have been traditionally considered stable through time. However, in highly dynamic systems with varying local environmental conditions and patch connectivity, communities are likely to experience temporal shifts in their assembly mechanisms. Here, we used a set of perennial (PR) and intermittent (IR) rivers to assess if assembly mechanisms vary seasonally in response to flow intermittence.

Location: Mediterranean climate region (100,000 km²), Spain.

Methods: We used a modelling approach to assess the relative effect of environmental sorting and dispersal-based processes on aquatic invertebrate metacommunities within and across river types at four distinct hydrological periods. We used local environmental variables to assess environmental sorting, and considered geographical, network and topographical distances as different dispersal surrogates. Linear mixed effect models accounting for the non-independence of pairwise distances were used to assess the relationships between community dissimilarity and distance matrices.

Results: Assembly mechanisms were more temporally stable in PR than in IR. In PR, community dissimilarities were equally related to environmental and geographical distances suggesting codominance of species sorting and dispersal-based assembly mechanisms. In IR, environmental distance best explained community dissimilarities during the dry period when flow cessation imposes strong environmental sorting, whereas metacommunity organization was much more stochastic during the rewetting period when high flows may randomly reorganize communities. Dispersal processes dominated assembly mechanisms between PR and IR during the rewetting period suggesting an increase in recolonization processes linking both river types following the dry period. Geographical and topographical distances best explained community variability, suggesting that overland dispersal dominates in river networks fragmented by drying events.

Main conclusions: Aquatic invertebrate metacommunity assembly mechanisms vary seasonally in response to changes in hydrological conditions. The temporal dimension should be better incorporated into metacommunity studies in highly dynamic systems such as intermittent rivers.

KEYWORDS

assembly mechanisms, beta diversity, dispersal, hydrological connectivity, macroinvertebrates, metacommunity, stream, temporal variability



1 | INTRODUCTION

Metacommunities are traditionally defined as an ensemble of local communities interconnected by the dispersal of multiple possibly interacting species (Leibold et al., 2004). Understanding how local communities are shaped and interact in a landscape is among the basic tenets of metacommunity studies. Assembly of metacommunities results from a combination of local factors (i.e. abiotic factors and species interactions), dispersal-based factors (species' dispersal abilities, spatial arrangement and connectivity of habitats) and stochastic events (extinction, speciation). Based on these assembly mechanisms, four main models have been commonly used to classify metacommunities: neutral, species sorting, mass effects and patch dynamics (Leibold et al., 2004). However, recent studies suggest that rather than four models with discrete boundaries, metacommunities should be described as a continuum shaped by different factors (i.e. environmental sorting, dispersal and stochastic processes) that vary in their relative importance (Winegardner, Jones, Ng, Siqueira & Cottenie, 2012). Most metacommunity studies so far have assumed that assembly mechanisms are relatively stable through time. However, metacommunities in highly dynamic systems (i.e. with varying local environmental conditions and patch connectivity) are very likely to experience frequent shifts in their organization and assembly mechanisms (Datry, Bonada & Heino, 2016). Although dynamic systems are common in nature (e.g. tidal zones, floodplains, fire-prone ecosystems, intermittent streams), ecologists are just starting to appreciate the temporal variability in their metacommunity assembly mechanisms (Fernandes, Henriques-Silva, Penha, Zuanon & Peres-Neto, 2014; Langenheder, Berga, Östman & Székely, 2012; Ruhí, Datry & Sabo, in press; Tonkin, Stoll, Jähnig & Haase, 2016).

Intermittent rivers (IR) are ideal systems to study how temporal changes in local environmental conditions and network connectivity can affect the relative roles of environmental sorting, dispersal and stochastic processes in shaping metacommunities (Cañedo-Argüelles et al., 2015; Datry, Bonada, et al., 2016). IR are characterized by alternating water flow and drying phases throughout the year, leading to shifts between lotic, lentic and terrestrial habitats that have a direct effect on local communities (Bogan & Lytle, 2007; Cid et al., 2015; Datry et al., 2014) and their connectivity (Bogan, Boersma & Lytle, 2013; Cañedo-Argüelles et al., 2015). The hydrological variability in IR is driven by seasonal climate variability and can be divided into four different periods: drying, dry, rewetting and wet. During the drying period, IR riffles (i.e. shallow and fast flowing river sections) start to dry and lose their connectivity but wet habitats remain numerous, whereas during the dry period IR have no discharge and wet habitats, if present, occur as isolated pools. Rewetting is characterized by increasing flow that reconnects riffles and the wet period by stable discharges allowing complete hydraulic connectivity of riffles and pools. On the contrary, perennial rivers (PR) offer relatively stable environmental conditions and permanent flow connectivity throughout the year, with some slight variations in flow characteristics during the rewetting (floods) and drying (low flows) periods. PR

and IR aquatic invertebrate communities differ in their composition (Bogan et al., 2013; Datry, Moya, Zubietta & Oberdorff, 2016; Schriever et al., 2015) and overall assembly processes (Datry, Moya, et al., 2016; Warfe et al., 2013). Because community variability and assembly processes depend on environmental heterogeneity (Astorga et al., 2014) and connectivity (Sarremejane, Mykrä, Bonada, Aroviita & Muotka, 2017), the relative importance of dispersal and environmental sorting in structuring IR metacommunities should change seasonally in response to variations in environmental conditions and flow connectivity (Datry, Bonada, et al., 2016). Contrarily, assembly mechanisms should be relatively stable in PR and environmental sorting should be the main driving force structuring their metacommunities (Grönroos et al., 2013; Warfe et al., 2013). PR and IR often co-occur within the same geographical area and PR are expected to serve as aquatic refuges for various species, being an important source of colonizers for IR after the dry period (Bogan & Boersma, 2012; Chester & Robson, 2011; Corti & Datry, 2016). However, no study to date has attempted to determine the community assembly mechanisms linking these two river types.

In this study, we assessed temporal variation in aquatic metacommunities in response to flow intermittency, using aquatic invertebrates in different mediterranean IR and PR as model organisms. First, we compared community assembly mechanisms (i.e. environmental sorting, dispersal and stochastic processes) within PR and IR along a climate-related seasonal gradient of hydrological conditions and network connectivity (i.e. drying, dry, rewetting and wet periods). Secondly, we assessed assembly mechanisms across both river types (PR-IR) to determine which mechanisms link their communities. Finally, we compared taxonomic and functional composition to better understand the mechanisms underlying community assembly processes. We expected (1) taxonomic and functional community composition and assembly mechanisms to be more temporally variable in IR than in PR, as flow intermittence in IR leads to considerable temporal changes in habitat conditions and connectivity that may strongly affect the stability of aquatic communities (e.g. Bêche & Resh, 2007). More specifically, we expected (2) the combined effect of environmental sorting and dispersal-based processes to constantly dominate metacommunity assembly in PR due to their relatively stable environmental conditions and high network connectivity facilitating species dispersal (Figure 1a). In IR, we expected a clear switch from dominance of environmental sorting during the drying, dry and wet periods to dominance of dispersal-based processes (mass effects) during the rewetting period (Figure 1b). Environmental sorting should be the main assembly mechanism in IR during the drying/dry period because increasing interspecific competition, changes from lotic to lentic conditions and increasing environmental variability among pools can act as strong filters selecting for the most adapted species (Chase, 2007; Datry, Bonada, et al., 2016). During the rewetting of IR, flood events should promote the recolonization of rewetted habitats through mass dispersal from surrounding refuges (Bogan & Boersma, 2012). Finally, during the wet period, environmental filtering should become gradually more important as

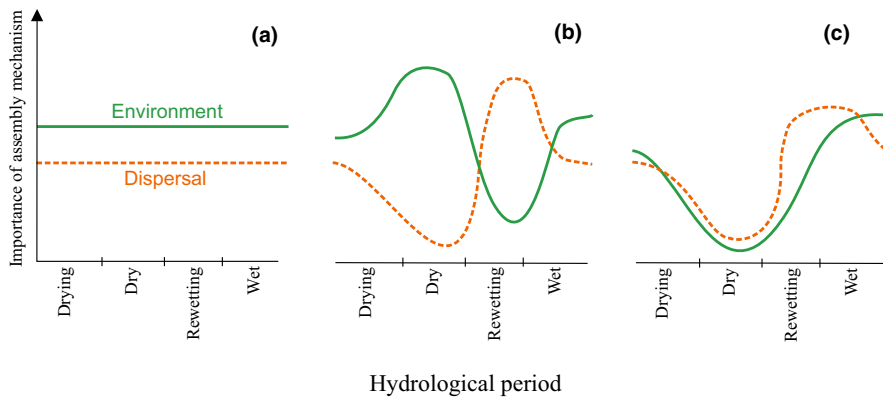


FIGURE 1 Expected effects of dispersal processes and environmental sorting on metacommunity assembly (y-axis) across hydrological periods (x-axis) in perennial rivers (a), intermittent rivers (b) and between the two river types (c) [Colour figure can be viewed at wileyonlinelibrary.com]

aquatic organisms get sorted to their most suitable habitats (Datry, Bonada, et al., 2016). Across river types, we expected (3) dispersal and environmental filtering to simultaneously drive community assembly between PR and IR, with the importance of both processes increasing during the rewetting and wet periods, when connectivity among sites and habitat homogeneity are highest at the landscape scale (Figure 1c). Contrarily, both environmental sorting and dispersal processes should have a low effect on community variability during the dry period as exchanges of species between PR and IR may be limited by the dryness of the landscape (leading to network fragmentation) and strong differences in habitat characteristics (Figure 1c).

2 | MATERIALS AND METHODS

2.1 | Biological and environmental data

We used seasonal biological and environmental data collected between April 1999 and February 2000 in 140 sites from 10 river basins in Spain (Bonada, Zamora-Muñoz, Rieradevall & Prat, 2005). Sites were sampled seasonally, i.e. on four different occasions, except the most ephemeral sites, which were dry for most of the year. Aquatic invertebrate samples consisted of kick-net samples (mesh size: 250 μ m) collected at each site from all available habitats until no new families were visually recorded after two successive samples (Bonada et al., 2005; Jáimez-Cuellar et al., 2002). Samples were further identified in the laboratory and taxa rank abundances were recorded as: 1 (1–3 individuals), 2 (4–10 individuals), 3 (11–100 individuals) and 4 (more than 100 individuals). Family-level taxonomic resolution has been shown to be sufficient in detecting community patterns (Bonada, Rieradevall, Prat & Resh, 2006; Rolls, Heino & Chessman, 2016) and should be an adequate surrogate for species- or genus-level resolution (Heino & Soininen, 2007). However, we tested for correspondence among community patterns at family and genus levels using an independent dataset (Appendix S1).

Environmental parameters measured at each site included phosphorus concentration (P-PO_4^{3-} mg/L), total nitrogen ($\text{NH}_4^+ + \text{N-NO}_2^- + \text{N-NO}_3^-$ mg/L), dissolved oxygen (mg/L), suspended solids (mg/L), electrical conductivity ($\mu\text{S/cm}$), pH, water temperature ($^\circ\text{C}$) and stream flow (l s). The index of riparian habitat quality (QBR, Munné, Prat, Solà, Bonada & Rieradevall, 2003) and the fluvial

habitat index (IHF, Pardo et al., 2002) were also recorded to assess local habitat heterogeneity. More information about how environmental parameters were measured can be found in Bonada et al. (2005). In addition, catchment area above the sampling site and site elevation were obtained using the geographical information system (GIS) software ARCGIS 10.0 and a 200-m Digital Elevation Model (DEM, Instituto Geográfico Nacional).

2.2 | Site selection

Only reference sites or sites with good ecological status were selected. The criteria used to discern between reference/good and non-reference sites were the presence of Trichoptera, which tend to be sensitive to pollution (Resh, 1993), and a value of the Iberian biotic index (IBMWP) higher than 71 in IR (Alba Tercedor et al., 2002; Sanchez-Montoya, Vidal-Abarca & Suárez, 2010). Sites were grouped as PR or IR based on discharge characteristics. Sites with discharge lower than 1 L/s on at least one sampling occasion (i.e. no flow or isolated pools) were classified as IR. As drying and rewetting in mediterranean rivers is highly seasonal (Bonada & Resh, 2013), samples were classified to four consecutive hydrological periods: drying (April–May), dry (July–August), rewetting (October–December) and wet (February–March). To avoid a varying number of sites among hydrological periods, we only considered sites that had aquatic invertebrate data for every period. The final set of sites included 39 PR and 17 IR (Figure 2). Differences in discharge between PR and IR among hydrological periods were tested using analysis of variance (ANOVA).

2.3 | Functional data

We used biological traits to determine if functional community composition differed among IR and PR and among hydrological periods. We collected six traits (24 categories) related to flow intermittence (e.g. Bonada, Rieradevall & Prat, 2007): organism size, life cycle duration, number of life cycles per year, dispersal strategy, resistance form and respiratory system (Tachet, Richoux, Bournaud & Usseglio-Polatera, 2010) (see Table S2.1 for a more detailed description). Trait information was compiled at family taxonomic level by averaging trait values of all genera in a given family. A fuzzy coding approach

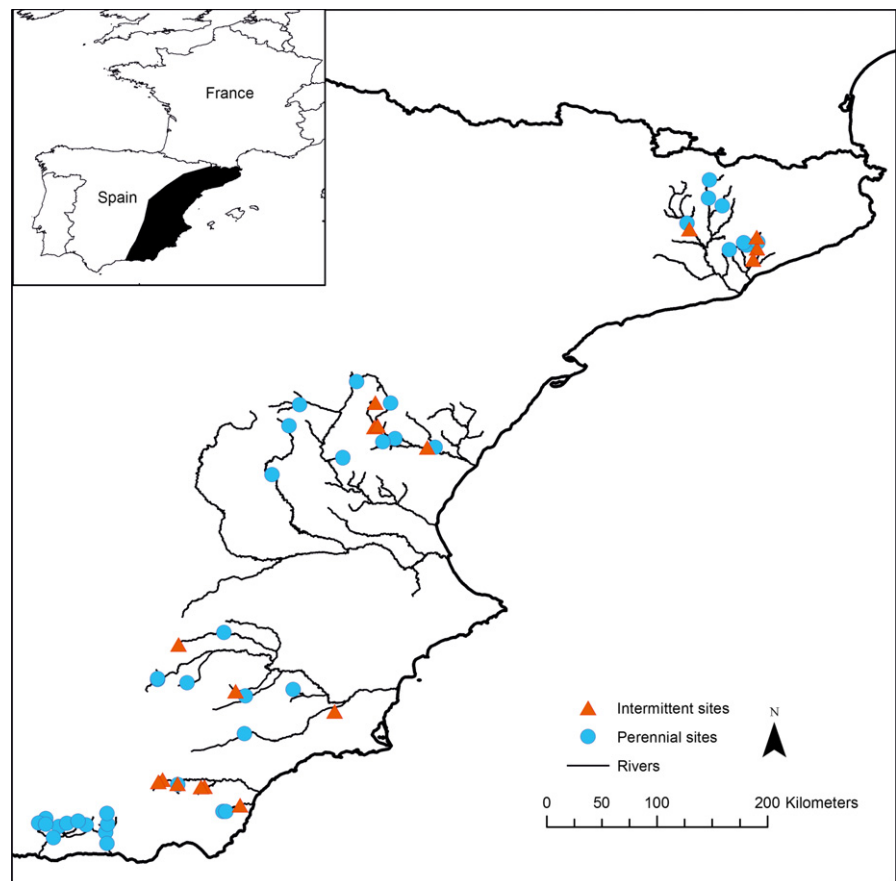


FIGURE 2 Site locations in the mediterranean climatic area of Spain. The black zone in the top-left map locates the study area [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jbi.13077)]

was used and affinities for each trait category were standardized as percentage affinities within a trait. We multiplied the taxa-trait matrix by the taxa-site abundance matrix to obtain a trait-site matrix that included the relative abundance of each trait category in each site.

2.4 | Distance matrices

We assessed the temporal variability in metacommunity assembly mechanisms by comparing the relationships between different distances (environmental, geographical, network, topographical) and community dissimilarity at each hydrological period among PR, among IR and between both river types (PR-IR). We used the environmental distance to assess environmental sorting, and considered geographical, network and topographical distances as different dispersal surrogates. We used the Sørensen dissimilarity index calculated on invertebrate presence-absence data to assess community variability (β -diversity, Anderson et al., 2011).

Prior to calculating the environmental distance, we used BIO-ENV analysis (Clarke & Ainsworth, 1993) to select the subset of environmental variables (excluding stream flow) most strongly correlated with the community dissimilarity matrix in each hydrological period across the river types. This procedure allowed us to avoid poor predictors that increase random variation in environmental distance calculation. The number of selected variables ranged from 3 (drying) to 8 (wet), depending on the period (Table S2.2).

Environmental distance was then calculated for each hydrological period as the Euclidean distance of standardized environmental variables between each pair of sites.

The dispersal-based distances included two physical distances (geographical and network distance) and a landscape resistance distance (topographical distance), which were used to approximate different potential pathways used by organisms to disperse between sites (Cañedo-Argüelles et al., 2015). The geographical distance considers the straight line between sites and was calculated as the Euclidean distance between geographical coordinates. The network distance measures the minimum distance between pairs of sites by following the river network. It was measured using a digitized stream network map (ECRIN database *EcrRiv.mdb*) and the Network Analyst extension in ARCGIS 10.0. However, because our sites were located in different catchments we were unable to calculate network distance between all the sites. The topographical distance considers dispersal through the landscape taking into account the topography and was calculated using a 200-m DEM, (IGN) and the program CIRCUITS-CAPE (McRae, 2006). CIRCUITS-CAPE uses electronic circuit theory (McRae, Dickson, Keitt & Shah, 2008) to calculate the resistance of the landscape to movements between each pair of sites based on map pixel values (elevation in our case). We hypothesized that flying organisms follow more easily the concavity of the landscape (valleys) than convex areas (e.g. hill slopes). Thus, pixels with high elevation values offer a higher resistance to organism movements than do those with low values. The pairwise resistance was then calculated as



the sum of the resistance of individual pixels taking into consideration all possible pathways across the landscape simultaneously. While geographical and topographical distances represented large-scale processes, network distance (only measured within the same network) characterized more local dispersal along the river network. Thus, a significant effect of geographical and topographical distances can be interpreted to result from dispersal limitation processes (distance between sites being too large for each species to reach each site at this scale) whereas a strong effect of network distance may reflect mass effects or dispersal limitation (source-sink dynamics) at the network scale depending on the connectivity of the network.

We performed tests of homogeneity of dispersion (PERMDISP, Anderson, Ellingsen & McArdle, 2006) using the Euclidean distances of standardized environmental variables (i.e. BIO-ENV selected subsets, excluding stream flow) and the geographical distance to ensure that similar environmental heterogeneity and spatial distribution existed among river types.

2.5 | Structural and functional community analyses

Differences in taxonomic and functional community composition between PR and IR and among hydrological periods were tested using nonparametric permutational multivariate analysis of variances (PERMANOVA on, permutation = 999). In addition, we performed PERMDISP to analyse if taxonomic variability (β -diversity: Sørensen dissimilarity) and trait variability (Gower distance) differed among river types and among hydrological periods.

We performed indicator species analysis (IndVal, Dufrene & Legendre, 1997) to identify indicator taxa and trait categories for each river type (i.e. PR and IR across all hydrological periods) and each hydrological period within a river type (i.e. drying, dry, rewetting and wet in PR and IR). This analysis calculates an indicator value for each taxa (or trait category) based on their frequency (or abundance) in each a priori determined group. Indicator values are constrained between 0 and 1 (1 indicating that a taxon is present at each site of a group) and their significance is tested with random permutations (here 999).

2.6 | Distance matrix processing and linear mixed effect model

As we were interested in assessing the effect of different distances (i.e. environmental and dispersal based distances) on community dissimilarity in IR and PR, and between both groups separately, we subdivided the pairwise community dissimilarity matrix in three submatrices. Each matrix included pairwise comparisons among IR sites, among PR sites and between PR-IR sites. The submatrices were built by replacing the disregarded pairwise comparison by NA's, e.g., for the between site submatrix, all within PR or IR sites comparisons were scored as NA's. This procedure was applied to each distance matrix and for each hydrological period. Also, we assumed that exchange of individuals between the most distant sites (751 km) was very unlikely at a short time-scale. Hence, we limited

our pairwise comparisons to certain distances. We thus determined the tipping point at which community dissimilarity ceased to increase with geographical distance (i.e. taxa turnover was no longer related with spatial factors). To determine the distance limit for pairwise comparisons, we fitted a local regression (LOESS) between Sørensen and geographical distances, separately for each hydrological period (Appendix S2: Figure S2.1), and identified the geographical distance at which the regression curve reached the asymptote ($197 \text{ km} < x < 346 \text{ km}$). We used among-period average geographical distance (263 km) as the distance limit for pairwise comparisons. Although most invertebrates are unlikely to disperse distances over 1 km (Poff et al., 2006), some migratory Odonata are able to travel several hundreds of kilometres (Bilton, Freeland & Okamura, 2001; Wikelski et al., 2006). In all distance matrices, all pairs of sites separated by a geographical distance exceeding 263 km were excluded from further analyses. The final PR, IR, and between PR-IR sites submatrices contained 58, 330, and 297 pairwise comparisons, respectively, except for the network distance that had a lower number of comparisons (22, 78 and 45, respectively).

We used linear mixed effect models for each submatrix and hydrological period to assess the relationship between community dissimilarity and each distance. Following recent research in landscape genetics (e.g. Phillipsen et al., 2015; Van Strien, Keller & Holderegger, 2012), we used minimum likelihood population effect (MLPE, Clarke, Rothery & Raybould, 2002) to account for the non-independence of the distance matrices. In this method, the random effect of the model incorporates a parameter (p) that accounts for the proportion of the total variance due to the correlation between distances that originates from the same sampling sites. We used restricted maximum likelihood (REML) and centred the explanatory variables around their mean to obtain unbiased estimates of the MLPE model parameters (Clarke et al., 2002; Van Strien et al., 2012). We assessed the amount of variation explained by the fixed effect (i.e., environmental distance, physical distances or resistance), using a marginal R^2 estimate for each model. For this, we used the R^2_{β} described in Edwards, Muller, Wolfinger, Qaqish and Schabenberger (2008). The R^2_{β} is calculated from the F -statistic and the degrees of freedom derived from Kenward-Roger's estimates using the difference in explained variation between the model that includes the fixed and random effects and a null model that only includes random effects. We used the R^2_{β} to compare the relative importance of the different distances in explaining community variability among river types and hydrological periods. Collinearity between predictor variables can be a serious problem when one wants to assess the "true" single effect of a variable (Dormann et al., 2013). We thus first tested if environmental and dispersal-based distances were correlated using a permutation ($n = 1,000$) test of the Pearson correlation between distance matrices. Then, in case of correlations between predictor variables (i.e. environmental, geographical, topographical and network distance) we calculated the R^2_{β} by comparing a model including the predictor of interest, its collinear variables and the random effects to a null model including only the collinear variables and the random effects. Because the network



distance included fewer pairwise comparisons than other distances we did not control for its collinear effect on other predictor distances that were calculated at a larger scale, although the single effect of the network distance was assessed including all possible collinear variables.

We conducted all statistical analyses in R (R Core Team, 2014) using the packages “vegan” (Oksanen et al., 2015) for the BIO-ENV, PERMDISP and PERMANOVA analyses, “ade4” (Chessel, Dufour & Thioulouse, 2004) for the functional analyses, “indicpecies” (De Cáceres & Legendre, 2009) for the indicator species analyses, “lme4” (Bates, Mächler, Bolker & Walker, 2014) for all MLPE models using lmer function, and “pbkrtest” (Halekoh & Højsgaard, 2014) for the Kenward-Roger’s estimates.

3 | RESULTS

3.1 | Environmental and spatial characteristics

Mean annual discharges were significantly lower in IR than in PR (ANOVA $F_{1,54} = 7.514$, $p = .008$) even though catchment areas were not significantly different ($F_{1,54} = 0.405$, $p = .527$). Discharge did not differ significantly among periods either in PR ($F_{3,152} = 0.708$, $p = .549$, Figure S2.2) or IR, although ANOVA bordered at significance in the latter ($F_{3,64} = 2.488$, $p = .068$). PR and IR were evenly distributed in space (PERMDISP on geographical distance: $F_{1,54} = 1.609$; $p = .210$) and did not differ in environmental heterogeneity in any period (PERMDISP on each period’s environmental distances $F_{1,54} < 2.395$; $p > .128$).

3.2 | Diversity and community composition

The average seasonal family richness (α) at a site ranged from 23 to 25 in PR and from 19 to 23 in IR, being the lowest during the rewetting period in both river types (Figure S2.3). The average seasonal β -diversity was generally higher in IR than in PR (Figure S2.3)

but did not differ significantly neither among hydrological periods nor between river types for any hydrological period (PERMDISP, Table 1). Similarly, functional variability did not differ among hydrological periods but was higher in IR than in PR during the wet period (Table 1). Taxonomic and functional composition were not significantly different among hydrological periods in IR but did differ significantly among hydrological periods in PR (Table 1). Taxonomic composition between PR and IR was significantly different during drying, dry and rewetting periods, and bordered at significance for the wet period, whereas trait composition differed significantly during the drying and rewetting period only (PERMANOVA, Table 1).

According to IndVal, several Trichoptera and Ephemeroptera families were significantly associated with PR, whereas several Odonata and Coleoptera were significantly associated with IR (Table 2). None of the taxa were significantly associated to any specific hydrological period in IR, whereas in PR most hydrological periods had significant indicator taxa (Table 2), e.g. communities during the dry period were dominated by several Heteroptera families. Regarding biological traits, drying resistance (e.g. respiration through spiracles, diapause) and resilience (aerial active dispersal, multivoltinism) were significantly associated with IR (Table 2), whereas flow adaptations (gills, without resistance forms, aquatic passive dispersal) and K-strategies (semivoltinism with long life cycles) were associated with PR (Table 2). Few traits were significantly associated with a given hydrological period (Table 2), e.g. aerial passive and multivoltinism dominated in IR during the rewetting period.

3.3 | Assembly mechanisms

Whereas most dispersal-based distances were significantly correlated with each other, the correlations between environmental and dispersal-based distances varied depending on the hydrological period and river type (Table S2.3).

In PR, the strength of the relationship between community dissimilarity and environmental, geographical and topographical

TABLE 1 Tests of homogeneity of dispersion (PERMDISP testing difference in community variability and functional variability) and nonparametric permutational multivariate analysis of variances (PERMANOVA testing differences in taxonomic and functional community composition) results among hydrological periods within perennial (PR) and intermittent rivers (IR) from the mediterranean climatic zone of Spain and between river types at each period (i.e. dry, drying, rewetting, wet). Sørensen dissimilarity and Gower distance were used as measures of taxonomic and functional dissimilarity respectively. p -values $< .05$ are indicated in bold

	Among periods				Between river types							
	IR		PR		Drying		Dry		Rewetting		Wet	
	F	p	F	p	F	p	F	p	F	p	F	p
Degrees of freedom	3, 64		3, 152		1, 54		1, 54		1, 54		1, 54	
<i>Taxonomic structure</i>												
PERMDISP	0.800	.500	2.232	.087	2.796	.100	1.830	.182	2.034	.160	2.551	.116
PERMANOVA	0.953	.505	1.709	.012	2.121	.012	2.602	.008	3.048	.004	1.824	.057
<i>Functional structure</i>												
PERMDISP	0.816	.489	0.860	.464	0.257	.615	2.410	.126	1.921	.171	7.162	.010
PERMANOVA	1.380	.151	2.083	.004	2.197	.040	1.377	.191	2.639	.016	1.169	.274



TABLE 2 Statistically significant ($p < .05$) indicator taxa and trait categories for perennial (PR) and intermittent rivers (IR) from the mediterranean climatic area of Spain and for each hydrological period within each river type (drying, dry, rewetting, wet). Indicator values (IV) are also reported

	IR			PR		
	Taxa					
	Family	IV	p	Family	IV	p
Between river types						
	Coenagrionidae	0.500	.001	Elmidae	0.735	.001
	Libellulidae	0.493	.003	Baetidae	0.723	.001
	Dryopidae	0.489	.042	Hydropsychidae	0.714	.001
	Haliplidae	0.488	.013	Rhyacophilidae	0.643	.001
	Glossiphoniidae	0.396	.004	Heptageniidae	0.639	.002
	Dolichopodidae	0.320	.019	Perlidae	0.576	.002
	Mesoveliidae	0.280	.039	Ephemerellidae	0.546	.001
				Psychodidae	0.513	.010
				Dixidae	0.449	.045
				Lepidostomatidae	0.446	.002
				Sericostomatidae	0.414	.016
				Ephemeridae	0.407	.007
Among hydrological periods						
Drying				Perlodidae	0.367	.035
				Chloroperlidae	0.340	.009
Dry				Notonectidae	0.425	.014
				Hydrometridae	0.404	.007
				Veliidae	0.384	.020
				Mesoveliidae	0.358	.003
Wet				Limnephilidae	0.513	.001
				Empididae	0.421	.014
				Capniidae	0.405	.002
Functional traits						
	Trait category	IV	P	Trait category	IV	P
Between river types						
	Respiration: spiracles (aerial)	0.736	.013	Number of generations: semivoltine	0.777	.001
	Number of generations: multivoltine	0.729	.002	Respiration: gills	0.725	.004
	Resistance form: diapause or dormancy	0.726	.026	Life cycle duration: >1 year	0.724	.014
	Dispersal: aerial active	0.718	.049	Resistance form: none	0.721	.003
	Life cycle duration: <1 year	0.716	.019	Dispersal: aquatic passive	0.720	.002
	Organism size: >8 cm	0.585	.004			
Among hydrological periods						
Drying				Organism size: 1–2 cm	0.516	.049
Dry	Life cycle duration: >1 year	0.543	.011	Respiration: spiracles (aerial)	0.561	.001
				Organism size: <0.25 cm	0.542	.030
				Life cycle duration: >1 year	0.526	.007
Rewetting	Number of generations: multivoltine	0.529	.043			
	Dispersal: aerial passive	0.542	.044			



distances was relatively stable among hydrological periods. The explanatory power of the environmental ($0.22 < R^2_{\beta} < 0.25$) and geographical ($0.23 < R^2_{\beta} < 0.32$) distances over community dissimilarity were almost equal, whereas the topographical distance ($0.02 < R^2_{\beta} < 0.06$) had the lowest explanatory power. The relationship between the network distance and PR metacommunity variability was relatively high during the drying period ($R^2_{\beta} = 0.21$) and gradually decreased towards the wet period ($R^2_{\beta} = 0.02$) (Figure 3a).

The relationship between the environmental distance and IR community dissimilarity was higher than or equal to that for PR during the drying ($R^2_{\beta} = 0.47$) and dry periods ($R^2_{\beta} = 0.21$), but R^2_{β} dropped during the rewetting period ($R^2_{\beta} = 0.11$) (Figure 3b). The effects of geographical and network distance on community dissimilarity were low at all hydrological periods ($R^2_{\beta} < 0.09$), whereas the effect of topographical distance was relatively high during the drying period, and it was the best explanatory distance during the wet period ($R^2_{\beta} = 0.14$).

The strength of the relationships between environmental and geographical distances and community dissimilarity across river types (PR-IR) were highest during the drying period ($R^2_{\beta} = 0.32$ and $R^2_{\beta} = 0.23$, respectively) and lowest during the dry period ($R^2_{\beta} = 0.14$ and $R^2_{\beta} = 0.13$ respectively) (Figure 3c). Network distance and topographical distance had generally low explanatory power over community dissimilarity, being highest during the wet ($R^2_{\beta} = 0.16$) and rewetting ($R^2_{\beta} = 0.12$) periods respectively.

4 | DISCUSSION

Our results show that the relative importance of different assembly mechanisms (environmental sorting, dispersal and stochastic processes) can vary through time in dynamic systems such as IR. As expected, metacommunity assembly mechanisms were more stable in PR than in IR, although community and functional trait composition were unexpectedly more temporally stable in the latter. As expected in PR, community dissimilarities were constantly related to environmental conditions and dispersal-based distances (Figures 1a and 3a). Although both drivers were important, dispersal was more important than environmental filtering in determining taxonomic turnover in PR. This is most likely related to the relatively homogeneous environmental conditions and high network connectivity in PR, which facilitate the assembly of metacommunities depending on taxon dispersal abilities. Environmental distance was clearly the strongest explanatory distance in IR during the drying and dry periods (Figure 3b), indicating the dominance of species sorting mechanisms. In contrast with our expectations, all distances had poor explanatory power over community dissimilarity in IR during the rewetting period, suggesting high stochasticity in community organization at this season. Finally, topographical distance was the best explanatory distance during the wet period, confirming the importance of dispersal-based processes among IR when the landscape is wet. These results thus imply that dispersal processes among IR might need a relatively long period of stable flow to be re-

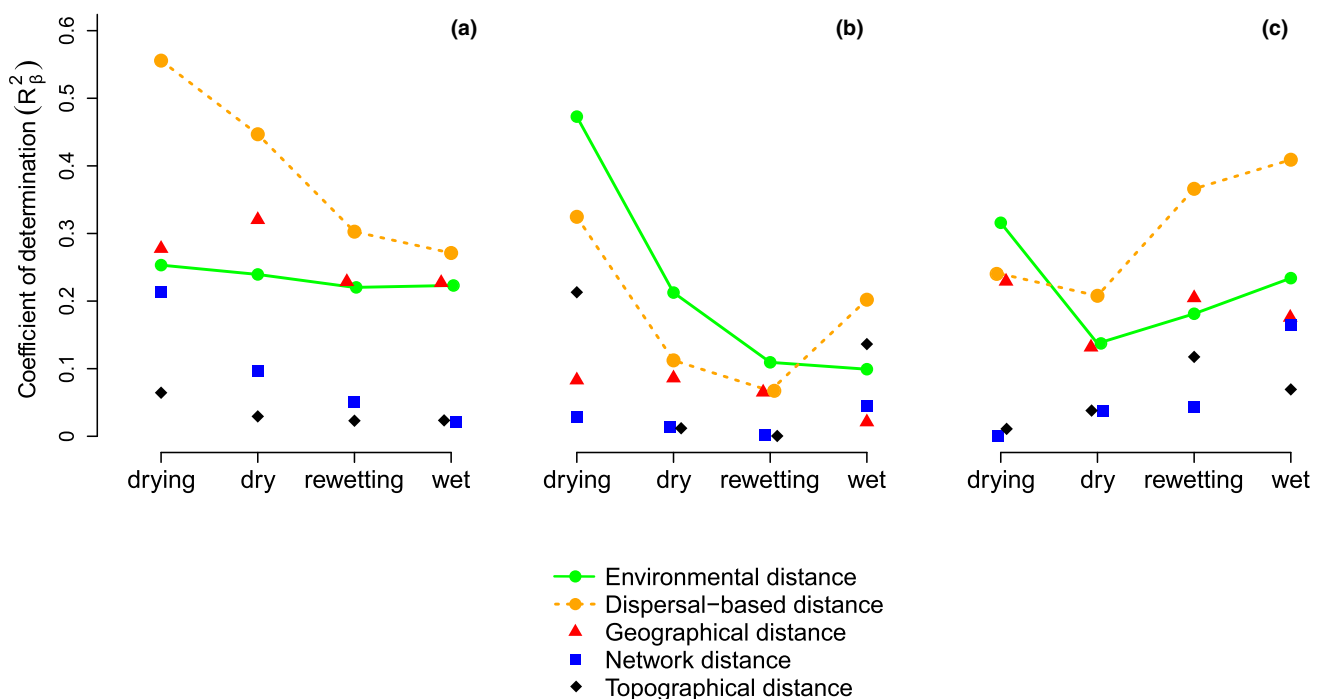


FIGURE 3 Strength of relationship between community dissimilarities and environmental, dispersal-based, geographical, topographic and network distances calculated using the coefficient of determination (R^2_{β} , y axis) for each hydrological period (x-axis) in perennial rivers (a), intermittent rivers (b) and across the river types (c). For comparison with Figure 1, only data points of environmental and dispersal-based distances are joined by a line. Dispersal-based distance is the sum of geographical, topographic and network distances [Colour figure can be viewed at wileyonlinelibrary.com]



established after network fragmentation by droughts. As expected, the explanatory power of dispersal-based distances (geographical and topographical) across river types (PR-IR, Figures 1c and 3c) increased during the rewetting period, indicating the dominance of dispersal-based processes during this period and a high importance of recolonization processes following the dry period.

Community persistence in highly dynamic systems mainly depends on the ability of species to cope with disturbances (e.g. ability for resistance or resilience). Frequently disturbed communities can be temporally more stable than communities under more benign conditions if organisms have traits that allow their survival and persistence under disturbed conditions (Scarsbrook, 2002). Hence, the stable community composition associated with variable assembly mechanisms in IR is most probably related to the capacity of organisms inhabiting those systems to tolerate changes in environmental conditions and hydrological connectivity by seasonally adapting their strategies (adaptation to drying and strong recolonization abilities) to respond to a natural disturbance regime (here drying/rewetting) (Bonada et al., 2007; Chester & Robson, 2011; Schriever et al., 2015). Boersma, Bogan, Henrichs and Lytle (2014) showed in a mesocosm experiment that species richness and community composition of arid-land aquatic invertebrates did not vary across drying intensity gradient, showing thus a strong resistance capacity of organisms inhabiting these intermittent systems. Moreover, the resistance of aquatic communities to seasonal drying in IR can be enhanced by the existence of hyporheic refugia, which are effectively used by the insect larvae (Del Rosario & Resh, 2000). Our functional trait analysis confirmed that IR communities have a higher incidence of traits related to disturbance adaptation such as short life-cycles with several potential reproductive cycles, active aerial dispersal and resistance forms. Conversely, in PR, taxonomic and functional community composition changed significantly across periods but always in response to similar mechanisms: environmental sorting and dispersal constraints. This can result from the fact that communities in stable environments are likely to comprise rare and sensitive taxa, whose presence can vary temporally in response to even small variations in hydrological conditions (Lepori & Malmqvist, 2009). Although PR are connected by flow throughout the year, they are subjected to seasonal hydrological changes, especially during the drying and rewetting periods, that can lead to changes in habitat condition and community composition (e.g. Huttunen, Mykrä & Muotka, 2012; Scarsbrook, 2002). Thus, low flows in summer and high flows in autumn may have caused local extinctions of the least adapted organisms, leading to significant modifications of community composition. PR may also be colonized by opportunistic taxa during the dry period: as flow is reduced and wet habitats disappear across the landscape, local competition may increase in remnant aquatic habitats and organisms adapted to low flows may thus be favoured.

Environmental sorting has been identified as the main driver of aquatic metacommunities when sampling is based on "snapshots" of relatively stable environments (i.e. perennial flows, Grönroos et al., 2013; Warfe et al., 2013). However, the effect of environmental sorting is likely to change seasonally if environmental conditions

change or if dispersal processes override their effect. The intensity of environmental sorting is expected to be stronger with increasing environmental harshness, since harsh conditions select for the most resistant/adapted taxa in the regional species pool (Chase, 2007). Although drying could be expected to homogenize metacommunities due to strong environmental filtering, we found that the relationship between environmental distance and community dissimilarity was strongest during the drying and dry periods in IR. This can be related to an increase in environmental heterogeneity among drying aquatic habitats and isolated pools (e.g. water temperature, pH, dissolved oxygen, Boulton [2003]) leading to stronger sorting along environmental gradients. Similarly, Rolls et al. (2016) showed that prolonged droughts can lead to high regional community variability by decreasing connectivity and increasing environmental heterogeneity. The weaker explanatory power of environmental distance in IR during the rewetting period may result from an increase in stochastic processes. Flow disturbances can promote stochasticity by causing local extinctions decoupled from environmental conditions and variability in post-disturbance recolonization (Campbell, Winterbourn, Cochrane & McIntosh, 2015; Lepori & Malmqvist, 2009). In our case the succession of drying and rewetting events could have promoted stochastic community organization among IR during the rewetting period, due to local extinctions (low local richness caused by severe drying and high flows) and random recolonization from surrounding communities by organisms with high reproductive and colonization capacities. Interestingly, environmental sorting between PR and IR was highest during the drying and wet periods, suggesting that when PR and IR have similar environmental conditions, their communities also tend to be more similar (Arscott, Larned, Scarsbrook & Lambert, 2010; Datry et al., 2014).

The effects of dispersal processes also varied in their intensity depending on the hydrological period and the river type. By affecting habitat and landscape connectivity, drying (increased pool isolation and disappearance of wet habitats across the landscape) and rewetting (rapid increase in flow connectivity and wet habitat coverage) are likely to affect dispersal processes. Overall, network distance had poor predictive power on community variability for IR and across the river types, suggesting that dispersal was not limited to river networks. Correspondingly, previous studies in rivers fragmented by droughts have found weak predictive power of network distance over metapopulation (Phillipsen & Lytle, 2013) and metacommunity structure (Cañedo-Argüelles et al., 2015). The river network has been shown to be an important dispersal pathway (Petersen, Masters, Hildrew & Ormerod, 2004), structuring invertebrate communities in fully connected networks (Brown & Swan, 2010; Sarremejane et al., 2017). However, overland dispersal among river basins may be even more important when the river network is partly dry (Cañedo-Argüelles et al., 2015; Razeng et al., 2016). The effect of dispersal should increase after a disturbance, allowing fast recolonization of disturbed patches (Datry, Bonada, et al., 2016). The high explanatory power of overland distances (topographical and geographical distances) on community variability across river types (PR-IR) in the rewetting period suggests that dispersal after drought occurred mainly overland.



Rewetting may have thus led to a mass spread of individuals from nearby perennial refuges towards IR, mainly through the concave landscape where wet habitats are more likely to be found. Previous studies have also reported that aquatic refugia (e.g. pools or perennial sections) distributed along the landscape can play a major role in the recolonization of IR after drought (Bogan & Boersma, 2012; Corti & Datry, 2016). Accordingly, the effect of topographical distance among IR was much higher during the drying and wet phases, suggesting that invertebrates mainly follow concavities of the landscape to travel among IR when the landscape is still wet. By contrast, the drastic decrease in the explanatory power of topographical distance from the drying to the dry period among IR communities indicates increased importance of stochastic processes and suggests that drying of the landscape may affect dispersal by decreasing the availability of wet pathways between rivers.

Our results show that metacommunity assembly mechanisms can vary temporally in highly dynamic systems such as intermittent river networks, even using such a coarse taxonomic level as family. Research at lower taxonomic levels (i.e. species or genus) could provide more profound information to better understand temporal variation in aquatic metacommunities in these systems. While traditional static metacommunity studies can be sufficient to the understanding of community assembly mechanisms in stable environments, temporal variability should be incorporated in studies dealing with dynamic systems. Although environmental sorting is often considered as the most important process structuring metacommunities, temporal changes in network and landscape connectivity may affect the exchange of species among patches and thereby enhance the relative importance of dispersal-based processes as drivers of metacommunity assembly.

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BIOSKETCH

Romain Sarremejane is a PhD candidate at the University of Oulu and his research focuses on the effects of connectivity and flow variability on aquatic communities and their assembly mechanisms. This study is the result of a research collaboration between Spanish and Finnish research groups initiated during his research visit at the University of Barcelona.

Author contributions: N.B. and N.P. collected the data; N.B., M.C.-A. and R.S. designed the study, R.S. conducted the statistical analysis with the help of M.C.-A; R.S. led the writing and other authors provided editorial advice.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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